# 3A Percentages, Percentage Change and GST

Standard Worksheet

Name: _____ Class: _____ Date: _____

## PART A · FLUENCY & SKILL PRACTICE

*Show complete step-by-step working in your workbook*

**Q1.** **Calculate each percentage of a quantity:**

(a) $15\%$ of $\$240$ (b) $35\%$ of $180\ \text{kg}$ (c) $4.5\%$ of $\$600$ (d) $125\%$ of $64\ \text{L}$

**Q2.** **Increase or decrease each quantity by the given percentage using a decimal multiplier:**

(a) Increase $\$80$ by $15\%$ (b) Decrease $\$150$ by $24\%$ (c) Increase $450\ \text{m}$ by $8\%$ (d) Decrease $\$320$ by $12.5\%$

**Q3.** **Calculate the percentage increase or decrease for each change:**

(a) From $\$50$ to $\$62$ (b) From $80\ \text{L}$ to $68\ \text{L}$ (c) From $\$125$ to $\$160$ (d) From $250\ \text{g}$ to $195\ \text{g}$

**Q4.** **Solve each Australian $10\%$ GST problem:**

(a) Pre-GST price is $\$140$; find the $10\%$ GST

(b) Pre-GST price is $\$260$; find the GST-inclusive price

(c) GST-inclusive price is $\$198$; find the pre-GST price

(d) GST-inclusive price is $\$462$; find the GST component

## PART B · PROBLEM-SOLVING & APPLICATIONS

*Write clear mathematical statements with units*

**Q5.** A plumber in Ballarat quotes $\$480$ (excluding GST) for replacing a hot-water valve. Calculate (a) the $10\%$ GST added to the invoice and (b) the total GST-inclusive invoice amount.

**Q6.** A receipt from a Melbourne hardware store shows a total GST-inclusive price of $\$319$ for a cordless drill kit. Find (a) the pre-GST price of the drill kit and (b) the exact amount of GST included in the receipt.

**Q7.** Water storage in Lake Eildon rose from $1250\ \text{GL}$ at the start of winter to $1475\ \text{GL}$ by the end of spring. Calculate the percentage increase in water storage.

**Q8.** After a $20\%$ member discount, a winter jacket at an outdoor store in Hobart costs $\$192$. What was the original full price of the jacket before the discount?

## PART C · REASONING & CHALLENGE

*Justify your conclusions using algebraic or geometric reasoning*

**Q9.** A bicycle's price is increased by $25\%$ from $\$400$ to $\$500$, and later decreased by $20\%$. Show by calculation that the final price returns to $\$400$, and explain using multipliers ($1.25 \times 0.80 = 1$) why a $25\%$ increase is undone by a $20\%$ decrease.

**Q10.** Explain why subtracting $10\%$ from a GST-inclusive price of $\$220$ gives $\$198$ instead of the true pre-GST price of $\$200$.